

TIC 220423668

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS

Period 3.651055 d (ALIASSED, see below)
-> corrected period: 7.302111 d
Epoch (t0) 1413.9498 BTJD
Duration 0.1200 d
Depth 4.525 % (45,247 ppm)
SNR 755.4

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

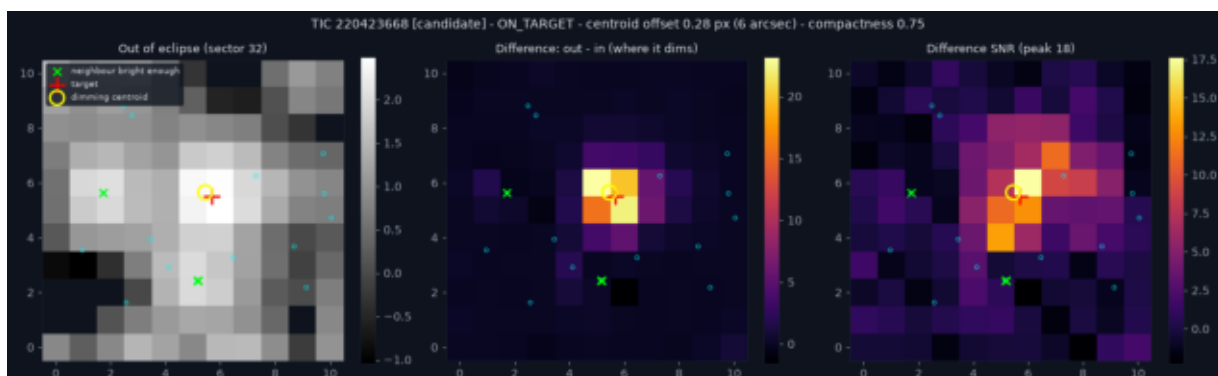
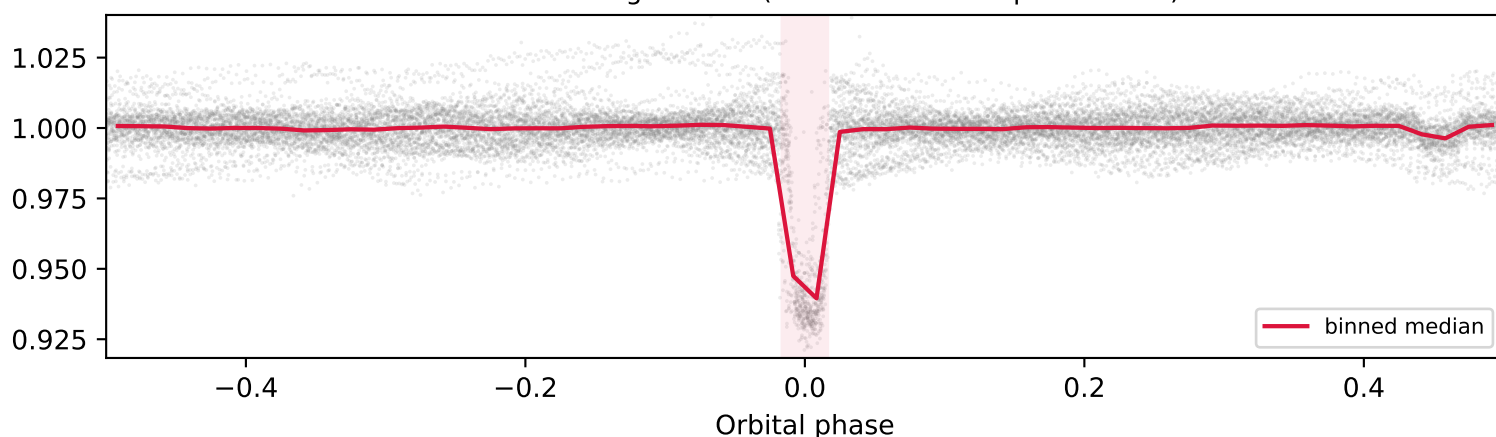
Eclipses observed 55 (gate ≥ 3)
Robust depth ratio 1.27 (gate ≥ 0.50)
Robust depth sigma 215.5 (gate ≥ 7.0)
Odd/even z-score 8.94 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 32 light-curve dip in this sector: 83.6 sigma t0 refined by 0.0025 phase
Difference-image peak SNR 17.67 (gate ≥ 5.0) Compactness 0.75 (gate ≥ 0.50)
Dimming centroid offset 0.28 px (6 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 220423674
3.23 px away, dTmag +1.58 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 8.94 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 7.302111 d, in any write-up.